

Chapter: 11**CHANGE OF STATE***Prepared by: KK Physics***Latent Heat**

It is the amount of heat required to change the state of unit mass of the substance from solid to liquid or from liquid to vapor without any change in temperature.

Since, amount of heat contained in a substance is directly proportional to the mass of the substance.

$$\text{i.e. } Q \propto m$$

$$\therefore Q = mL$$

where L is a proportionality constant and is called latent heat or hidden heat.

So, $L = \frac{Q}{m}$. Its SI unit is J kg^{-1} and CGS unit is Cal gm^{-1} .

There are two types of Latent Heat:

- a) Latent heat of Fusion
- b) Latent heat of Vaporization

We know that a solid changes into a liquid at a temperature equal to its melting point. Similarly, a liquid changes into gaseous state at a temperature equal to its boiling point. During heating the substance, the heat has been used up in changing the state. Temperature remaining the same till the state of the entire substance has changed. This energy is used up in separating the molecules farther apart and as work must be done against the surrounding pressure to change the state of a substance. That is why heat energy spent in change of state is called latent heat i.e. heat which is hidden from the thermometer.

Latent Heat of Fusion

It is defined as the amount of heat required to change the unit mass of substance from solid state to liquid state without any change in temperature. It is denoted by L_f . The latent heat of fusion of ice is 80 Cal gm^{-1} in CGS system or $3.36 \times 10^5 \text{ Jkg}^{-1}$ in SI system. If Q be the amount of heat required to convert a substance of a mass m and latent heat of fusion L_f .

$$\text{Then, } Q = m L_f$$

Latent Heat of Vaporization:

It is defined as the amount of heat required to change the unit mass of substance from liquid state to vapor state without any change in temperature. The latent heat of vaporization of water is 540 Cal gm^{-1} in CGS and $2.26 \times 10^6 \text{ J Kg}^{-1}$ in SI system. If Q be the heat required to convert a substance of a mass m and latent heat of vaporization is L_v

$$\text{Then, } Q = m L_v$$

Determination of Latent heat of Fusion of ice by the method of mixture

Principle:

This experiment is based on the principle of calorimetry. According to this, if there is no exchange of heat with the surrounding, heat gained by the cold body is equal to the heat lost by the hot body.

Procedure:

The latent heat of fusion of ice can be measured by the method of mixture as shown in fig. Let us consider a dry and clean calorimeter with stirrer and its weight is measured. It is filled almost half of its volume with warm water. The initial temperature of calorimeter and its contents be noted. Then some pieces of ice at 0°C are dropped in the calorimeter and the mixture is stirred until the ice is completely melted. Then the final temperature of the mixture is noted.

3/31/2025

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5

Calculation:

Let,

m_c = mass of calorimeter and stirrer

m_w = mass of water

S_c = specific heat capacity of calorimeter

S_w = specific heat capacity of water

m_i = mass of ice

L_f = latent heat of fusion of ice =?

θ_1 = initial temperature of calorimeter and water

θ = final temperature of the mixture

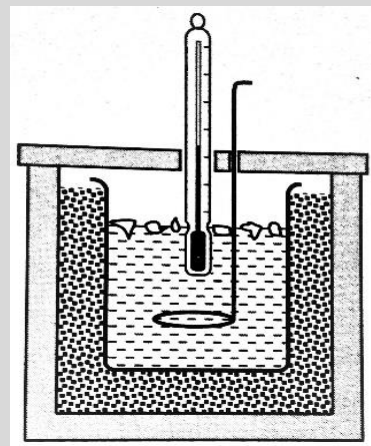


Fig. Measurement of latent heat of fusion of ice

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6

In this process the calorimeter, stirrer and water losses heat whereas ice gains heat.

Now, the amount of heat lost by the calorimeter and water from $\theta_1^\circ\text{C}$ to $\theta^\circ\text{C}$ is

$$\begin{aligned} Q_1 &= m_c S_c (\theta_1 - \theta) + m_w S_w (\theta_1 - \theta) \\ &= (m_c S_c + m_w S_w) (\theta_1 - \theta) \dots\dots\dots (i) \end{aligned}$$

Here, the amount of heat gained by ice is in two steps:

(i) Amount of heat gained by ice to change it from ice at 0°C to water at 0°C is $= m_i L_f$

(ii) Amount of heat gained by ice to change from water (formed from ice) at 0°C to mixture

at $\theta^\circ\text{C}$ is $= m_i S_w (\theta - 0) = m_i S_w \theta$

Thus, total amount of heat gained by ice $Q_2 = m_i L_f + m_i S_w \theta \dots\dots\dots (ii)$

According to principle of calorimetry,

Total heat gained = Total heat lost

$$\text{Or, } m_i L_f + m_i S_w \theta = (m_c S_c + m_w S_w) (\theta_1 - \theta)$$

$$\text{Or, } m_i L_f = (m_c S_c + m_w S_w) (\theta_1 - \theta) - m_i S_w \theta$$

$$\text{Or, } L_f = \frac{(m_c S_c + m_w S_w) (\theta_1 - \theta) - m_i S_w \theta}{m_i}$$

By knowing all the values in RHS, the latent heat of fusion of ice can be calculated.

Determination of Latent heat of vaporization of water by the method of mixture

Principle: This experiment is based on the principle of calorimetry. According to this, if there is no exchange of heat with the surrounding, heat gained by the cold body is equal to the heat lost by the hot body.

Procedure: The latent heat of vaporization of water can be measured by the method of mixture as shown in fig. Let us consider a dry and clean calorimeter with stirrer and its weight is measured. It is filled almost half of its volume with water. The initial temperature of calorimeter and its contents be noted. Then the steam is passed from the boiler to the calorimeter and the mixture is stirred. After some time, the temperature is increased to a constant value. Then, the final temperature of the mixture is noted.

3/31/2025

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9

Calculation:

Let,

m_c = mass of calorimeter and stirrer

m_w = mass of water

m_s = mass of steam

θ_1 = initial temperature of calorimeter and water.

θ_2 = temperature of steam = 100°C

θ = final temperature of the mixture

S_w = specific heat capacity of water

S_c = specific heat capacity of calorimeter

L_v = latent heat of vaporization = ?

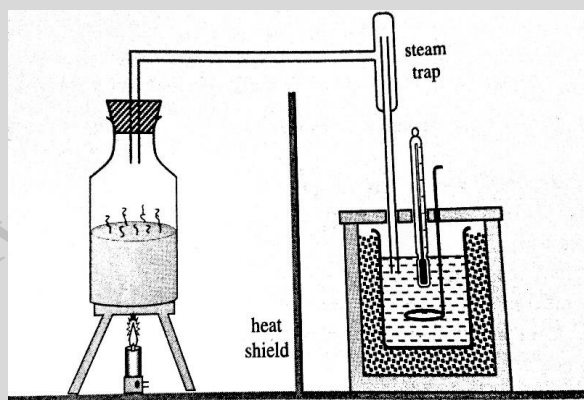


Fig. Measurement of latent heat of vaporization of water

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10

In this process, steam losses heat whereas calorimeter and water gains heat

Now, the amount of heat gained by the calorimeter and water form $\theta_1^\circ\text{C}$ to $\theta^\circ\text{C}$ is

$$\begin{aligned} Q_1 &= m_c S_c (\theta - \theta_1) + m_w S_w (\theta - \theta_1) \\ &= (m_c S_c + m_w S_w) (\theta - \theta_1) \dots\dots\dots (i) \end{aligned}$$

Here, the amount of heat lost by vapor is in two steps:

(i) Amount of heat lost by vapor in changing from vapor at 100°C to water at 100°C is $= m_s L_v$

(ii) Amount of heat lost by vapor in changing from water (formed from vapor) at 100°C to mixture at $\theta^\circ\text{C}$ is $= m_s S_w (100 - \theta)$

Thus, total amount of heat lost by vapor $Q_2 = m_s L_v + m_s S_w (100 - \theta) \dots\dots\dots (ii)$

According to principle of calorimetry,

Total heat lost = Total heat gained

$$\text{Or, } m_s L_v + m_s S_w (100 - \theta) = (m_c S_c + m_w S_w) (\theta - \theta_1)$$

$$\text{Or, } m_s L_v = (m_c S_c + m_w S_w) (\theta - \theta_1) - m_s S_w (100 - \theta)$$

$$\text{Or, } L_v = \frac{(m_c S_c + m_w S_w) (\theta - \theta_1)}{m_s} - S_w (100 - \theta)$$

By knowing all the values in RHS, the latent heat of vaporization of water can be calculated.

Effect of Pressure on Melting Point:

There are two kinds of substance which expands or contracts on melting. Most of the metals like copper, gold, silver etc. expand on melting while solids like rubber, ice, cast iron, bismuth etc. contract on melting.

In case of substance which expand on melting, the melting point of substance is increased on increasing the pressure and vice versa.

In case of substance which contract on melting, the melting point is decreased on increasing the pressure and vice versa.

Effect of pressure on boiling point:

The boiling point of a liquid increases with increase in pressure while decreases with decrease in pressure. Thus, water will boil at a temperature higher than 100°C , if the atmospheric pressure is higher than 760 mm and will boil at a lower temperature if the pressure is less than 760 mm of Hg. Pressure cooker is based on the principle that boiling point of liquid increases with increase in pressure. Thus, in pressure cooker, pressure is more than the atmosphere so boiling point increases and food cooks faster than in open vessel.

Explain the terms:

i) Evaporation: The slow and silent process of conversion of a liquid state to gaseous state at all temperature is called evaporation. When evaporation takes place, the molecules at the surface of liquid takes away some of heat energy from the remaining molecules. As a result, the heat energy of the remaining molecules get reduced. Therefore, cooling effect is seen in the case of evaporation.

It has been found that the rate of evaporation of a liquid depends upon:

- i. Nature of the liquid
- ii. Area of exposed surface of liquid.
- iii. Pressure and temperature of the atmosphere
- iv. Effect of wind
- v. Humidity of air

Application of Evaporation

- a. Doctor advises to put the strips of wet cloth on the forehead of a person having high temperature (fever).
- b. In hot season, water gets cooled in an earthen pot.
- c. Dogs keep their tongues out usually in summer.
- d. A person feels cooling when he pours a little sprit on his hand.
- e. It takes longer time to dry wet cloths in rainy summer than in sunny winter.
- f. We feel cold only while coming out in open air after having bath in a bathroom but not inside bathroom.
- g. We feel cold when electric fan is operated in a closed room.
- h. We feel cool in mouth when we eat halls.

ii) Boiling:

The rapid and noisy conversion of the liquid from its liquid state to vapor state at particular temperature is called boiling. During this process, temperature of the liquid does not change and no cooling effect is observed.

Differences between Evaporation and Boiling

Evaporation	Boiling
i) It is the slow and silent process of conversion of liquid into vapour.	i) It is the fast and noisy process of conversion of liquid into vapour.
ii) It takes place at all temperature.	ii) It takes place at a fixed temperature (boiling point).
iii) It takes place at the surface of the liquid.	iii) It take place throughout the volume of the liquid
iv) Cooling effect is observed.	iv) No cooling effect is observed.

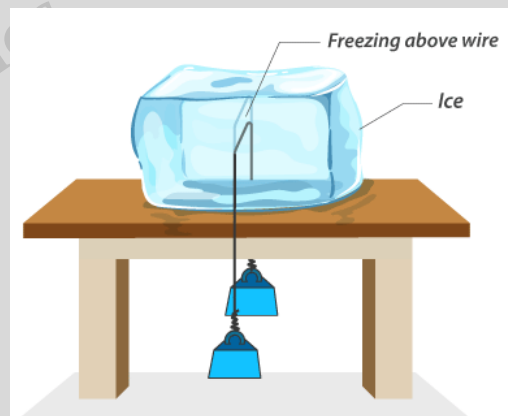
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17

iii) Super Cooling: When a liquid is cooled gently, experimentally it is found that the liquid may cool even several degrees below its freezing point without solidification. Such a liquid at a temperature below its freezing point is called super cooled liquid.

iv) Regelation: The process of melting of ice under increased pressure and again solidification after the removal of pressure is called regelation. For example:
 a) when two pieces of ice are pressed each other then they stick together forming a single piece. b) when a fine wire with two heavy loads is passed over a block of ice, the wire finds its way through the ice block without cutting it into two pieces.



3/31/2025

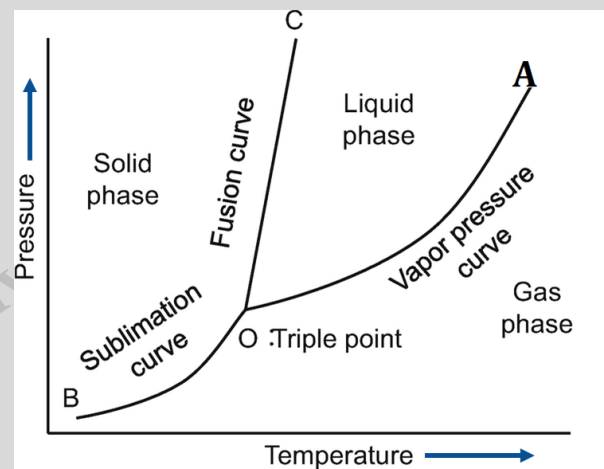
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18

Triple Point

It is a point in phase diagram which represents a particular value of pressure and temperature at which the solid, liquid and gaseous state of the substance can co-exist.

In other words, the triple point of a substance is the point in phase diagram at which the vaporization curve, fusion curve and sublimation curve intersect.



3/31/2025

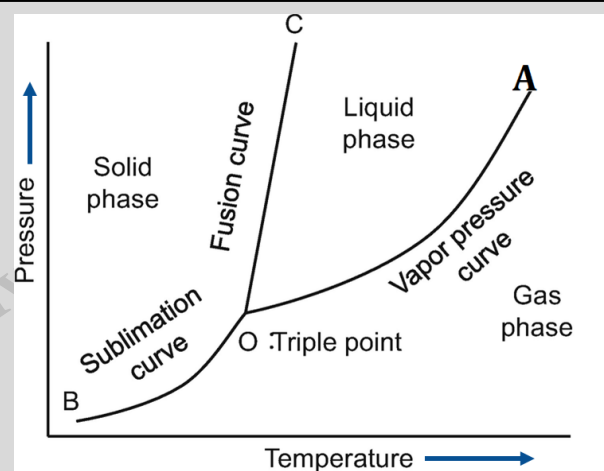
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19

The curve in a phase diagram, which distinguishes the solid state and liquid state of a substance is known as fusion line (fusion curve, OC).

The curve which separates the liquid state and gaseous state of a substance is known as vaporization line (vaporization curve, OA).

Similarly the curve which separates the solid state and gaseous state of a substance is known as sublimation line or sublimation curve, BO



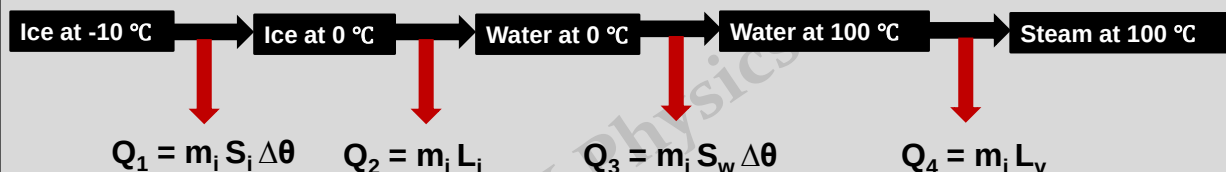
These three curves meet at point O.
Thus point O is the triple point.

3/31/2025

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20

1) Calculate the amount of heat required to change 10 gm of ice at -10°C into steam at 100°C . (Sp. Heat of ice is $0.5 \text{ Cal gm}^{-1} \text{ C}^{-1}$, latent heat of ice 80 Cal gm^{-1} , sp. Heat of water is $1 \text{ Cal gm}^{-1} \text{ C}^{-1}$, and latent heat of steam is 540 Cal gm^{-1} .)



So, Total Heat Required $Q = Q_1 + Q_2 + Q_3 + Q_4$

Here, mass remains constant
So, $m_i = m_w$

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21

Q.2) What is the result of mixing 20 gm of water at 60°C with 10 gm of ice at 0°C ?

Here, Mass of water (m_w) = 20 gm

Temperature of water (θ_1) = 60°C

Mass of ice (m_i) = 10 gm

let θ be the final temperature when the ice is mixed with water.

Heat lost by water when its temperature changes from 60°C to $\theta^{\circ}\text{C}$ is

$$\begin{aligned} Q_1 &= m_w S_w (60 - \theta) \\ &= 20 \times 1 \times (60 - \theta) \\ &= 1200 - 20\theta \dots\dots\dots(i) \end{aligned}$$

Heat gained by ice when it changes from ice at 0°C to water at 0°C and then to water at $\theta^{\circ}\text{C}$

$$\begin{aligned} Q_2 &= m_i L_i + m_i S_w (\theta - 0) \\ &= 10 \times 80 + 10 \times 1 \times \theta \\ &= 800 + 10\theta \dots\dots\dots(ii) \end{aligned}$$

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22

Q.3) What is the result of mixing 100 gm of ice at 0 °C and 20 gm of steam at 100 °C?

Heat gained by ice when it changes from ice at 0 °C to water at 0 °C and then water at θ °C is

$$Q_1 = m_i L_i + m_i S_w \Delta\theta = \dots\dots\dots$$

Heat lost by steam when it changes from steam at 100 °C to water at 100 °C and then water at θ °C

$$Q_2 = m_s L_v + m_s S_w \Delta\theta = \dots\dots\dots$$

according to principle of calorimetry

heat gained = heat lost

$$\theta = ?$$

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23

Q.4) What is the result of mixing 10 gm of ice at 0 °C and 100 gm of water at 60 °C?

Do yourself

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24

Q.5) find the result of mixing 10 gm of ice at - 8 °C with 10 gm of water at 12 °C. Sp. Heat capacity of ice is 0.5 cal gm⁻¹ °C⁻¹ and latent heat of ice is 80 cal gm⁻¹.

Here, mass of ice (m_i) = 10 gm at - 8 °C

mass of water (m_w) = 10 gm at 12 °C

sp.heat capacity of ice (S_i) = 0.5 cal gm⁻¹ °C⁻¹

latent heat of ice (L_i) = 80 cal gm⁻¹

Let θ be the final temperature of the mixture.

Now, heat required to convert ice from - 8 °C to 0 °C ice is

$$Q_1 = m_i S_i \Delta\theta = 10 \times 0.5 \times 8 = 40 \text{ cal}$$

Heat required to convert 0 °C ice to 0 °C water is

$$Q_2 = m_i L_i = 10 \times 80 = 800 \text{ cal}$$

Heat required to convert 0 °C water (formed from ice) to θ °C water is

$$Q_3 = m_i S_w \Delta\theta = 10 \times 1 \times (\theta - 0) = 10 \theta$$

So, total heat required (gained)= 40 + 800 + 10 θ = 840 + 10 θ

$$\Delta\theta = 0 - (-8) \\ = 8$$

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25

Heat given (lost) by water to convert from 12 °C to θ °C is

$$Q_4 = m_w S_w \Delta\theta = 10 \times 1 \times (12 - \theta) = 120 - 10\theta$$

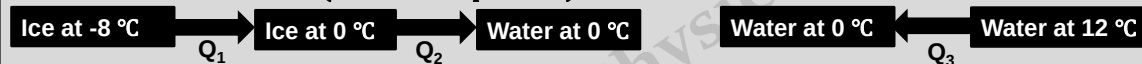
We have

Heat gained = heat lost

$$\text{or, } 840 + 10\theta = 120 - 10\theta$$

$$\text{or, } 20\theta = -720$$

$$\therefore \theta = -36^\circ\text{C (which is not possible)}$$



Here, $Q_1 = m_i S_i \Delta\theta = 40 \text{ cal}$, $Q_2 = m_i L_i = 800 \text{ cal}$ and $Q_3 = m_w S_w \Delta\theta = 120 \text{ cal}$

Here, $Q_1 + Q_2 > Q_3$

$\therefore$ total heat given by water is not sufficient to melt all ice.

Here, 40 cal heat is already used to convert - 8 °C ice to 0 °C ice.

So, remaining heat is 120 - 40 = 80 cal.

Now, Let this 80 cal heat is used to melt m gm of ice

$$\text{i.e. } mL_i = 80 \Rightarrow m = 1 \text{ gm}$$

Thus, 1 gm of ice is melted. So final temperature of mixture is 0 °C and it contains 9 gm of ice and 11 gm of water.

3/31/2025

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26

Q.6) What mass of water must evaporate from the surface of 80 kg human body to cool it by 1 °C. (Sp. heat capacity of human body is 0.1 cal gm⁻¹K⁻¹ and latent heat of vaporization of water at the body temperature is 577 cal g⁻¹)

Mass of human body (m_h) = 80 kg = 80,000 gm

Temperature difference ($\Delta\theta$) = 1 °C

Sp. heat capacity of human body (S_h) = 0.1 Cal gm⁻¹ K⁻¹

Latent heat of vaporization (L_v) = 577 Cal gm⁻¹

Mass of water evaporate (m_w) = ?

Now, heat lost by human body when his temperature decreases by 1 °C is

$$Q_1 = m_h S_h \Delta\theta = 80,000 \times 0.1 \times 1 = 8,000 \text{ cal}$$

So 8000 cal of heat is absorbed by water from the human body to evaporate.

Let m be mass of water which absorbs 8000 cal of heat

$$\text{So, } m_w L_v = 8000$$

$$\text{or, } m_w \times 577 = 8000$$

$$\text{or, } m_w = 13.865 \text{ gm}$$

3/31/2025

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27

Q.7) 10 gm of steam at 100 °C is passed into a mixture 100 gm of water and 5 gm of ice at 0 °C. Find the resulting temperature of the mixture.

Given, Mass of water (m_w) = 100gm at 0 °C

mass of ice (m_i) = 5 gm at 0 °C

mass of steam (m_s) = 10 gm at 100 °C

latent heat of ice (L_i) = 80 cal/gm

latent heat of steam (L_s) = 540 cal/gm

Let θ be the final temperature of the mixture.

Now, heat lost by steam when it changes from 100 °C steam to 100 °C water and then θ °C water is

$$\begin{aligned} Q_1 &= m_s L_s + m_s S_w (100 - \theta) \\ &= 10 \times 540 + 10 \times 1 \times (100 - \theta) \\ &= 5400 + 1000 - 10\theta \\ &= 6400 - 10\theta \end{aligned}$$

Heat gained by ice to change from 0 °C ice to 0 °C

$$\text{water } Q_2 = m_i L_i = 5 \times 80 = 400 \text{ cal}$$

Here, 5 gm ice is converted into 0 °C water. So,

total mass of water at 0 °C is 100 + 5 = 105 gm

Now, heat gained by 105 gm water when it changes from 0 °C water to θ °C water is

$$Q_3 = 105 \times 1 \times (\theta - 0) = 105\theta$$

$$\text{Total heat gained} = Q_2 + Q_3 = 400 + 105\theta$$

We have heat gained = heat lost

$$400 + 105\theta = 6400 - 10\theta$$

$$\text{or, } 115\theta = 6000$$

$$\text{or, } \theta = 52.17 \text{ °C}$$

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28

Q.8) What is the result of mixing when 10 gm of ice at -10°C is mixed with 100 gm of water at 90°C ?

Given: mass of ice (m_i) = 10 gm at -10°C

mass of water (m_w) = 100 gm at 90°C

latent heat of ice (L_i) = 80 cal/gm

Now, heat required by ice to convert -10°C ice to 0°C ice and then 0°C water is

$$Q_1 = m_i S_i \Delta\theta + m_i L_i = 10 \times 0.5 \times 10 + 10 \times 80 \\ = 850 \text{ cal}$$

Amount of heat given by water when its temperature changes from 90°C to 0°C is

$$Q_2 = m_w S_w \Delta\theta = 100 \times 1 \times 90 = 9000 \text{ cal}$$

Here 850 cal heat is used to melt 10 gm ice

then remaining heat is $9000 - 850 = 8150 \text{ cal}$

And both 110 gm (i.e. ice and water) are at 0°C water.

Now, remaining heat (i.e. 8150 cal) is used to increase the temperature of 110 gm water from 0°C to $\theta^{\circ}\text{C}$.

$$\therefore 8150 = 110 \times 1 \times (\theta - 0)$$

$$\theta = 74.09^{\circ}\text{C}$$

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29

Q.9) What is the result of mixing when 100 gm of ice at -10°C is mixed with 80 gm of water at 50°C ?



Given: mass of ice (m_i) = 100 gm at -10°C

mass of water (m_w) = 80 gm at 50°C

latent heat of ice (L_i) = 80 cal/gm

Now, heat required by ice to convert -10°C ice to 0°C ice is

$$Q_1 = m_i S_i \Delta\theta = 100 \times 0.5 \times 10 = 500 \text{ cal}$$

Heat required to melt 100 gm ice is

$$Q_2 = m_i L_i = 100 \times 80 = 8000 \text{ cal}$$

$$\therefore \text{total heat required} = Q_1 + Q_2 = 8500 \text{ cal}$$

3/31/2025

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30

Amount of heat given by water when its temperature changes from 50 °C to 0 °C is

$$Q_3 = m_w S_w \Delta\theta = 80 \times 1 \times 50 = 4000 \text{ cal}$$

Here, the heat given by water (i.e. 4000 cal) is not sufficient to melt the ice completely.

∴ let only m gm of ice will melt.

But here 500 cal heat is already used to change the temperature of ice from -10 to 0 °C .

Thus, remaining heat = 4000 - 500 = 3500 cal. This heat is used to melt m gm of ice

$$\text{i.e. } mL_i = 3500$$

$$\text{or, } m \times 80 = 3500$$

$$\text{or, } m = 43.75 \text{ gm}$$

Now, total mass of water = 80 + 43.75 = 123.75 gm at 0 °C

$$\text{remaining mass of ice (un-melted)} = 100 - 43.75 = 56.25 \text{ gm at } 0^\circ\text{C}$$

3/31/2025

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31

Q.10) 500 gm of ice at -16 °C is dropped into a calorimeter containing 1000 gm of water at 20 °C. The calorimeter is of copper and has a mass of 278 g. compute the final temperature of the system. ($S_c = 400 \text{ Jkg}^{-1}\text{K}^{-1}$, $S_i = 2100 \text{ Jkg}^{-1}\text{K}^{-1}$ $L_i = 3.36 \times 10^5 \text{ Jkg}^{-1}$)

Heat required to change the temperature of ice

from - 16 °C to 0 °C is

$$Q_1 = m_i S_i \Delta\theta = 0.5 \times 2100 \times 16 = 16800 \text{ J}$$

Heat required to change ice at 0 °C to water at 0 °C is

$$Q_2 = m_i L_i = 0.5 \times 3.36 \times 10^5 = 168000 \text{ J}$$

So total heat required = $Q_1 + Q_2 = 184800 \text{ J}$

Total heat given by water and calorimeter when their temperature changes from 20 °C to 0 °C is

$$\begin{aligned} Q_3 &= m_w S_w \Delta\theta + m_c S_c \Delta\theta \\ &= 1 \times 4200 \times 20 + 0.278 \times 400 \times 20 \\ &= 86224 \text{ J} \end{aligned}$$

$$\text{Here } Q_1 + Q_2 > Q_3$$

So all ice cannot be melted.

Here, 16800 J is already used to change the temperature of ice from - 16 °C to 0 °C

$$\text{So remaining heat} = 86224 - 16800 = 69424 \text{ J}$$

Thus, 69424 J heat is used to melt m kg of ice

$$\therefore mL = 69424$$

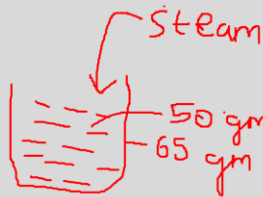
$$m = ?$$

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32

Q.11) A copper calorimeter weighing 65 gm contains 50 gm of water at 20 °C. When some steam is passed into water, the temperature of the calorimeter with its contents rose to 25 °C. Calculate the mass of steam condensed. ($S_c = 0.094 \text{ Cal gm}^{-1}\text{K}^{-1}$, $S_{L_v} = 540 \text{ cal gm}^{-1}$)



Heat gained by W & C = $m_s \Delta \theta + m_c \Delta \theta$

" lost by S = $m L_v + m S_w \Delta \theta$

3/31/2025

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33

Q.12) A mixture of 500 gm water and 100 gm ice at 0 °C is kept in a copper calorimeter of mass 200 gm. How much steam from the boiler be passed to the mixture so that the temperature of the mixture reaches to 40 °C ? ($S_c = 400 \text{ Jkg}^{-1}\text{K}^{-1}$, $S_w = 4200 \text{ Jkg}^{-1}\text{K}^{-1}$, $L_i = 3.36 \times 10^5 \text{ Jkg}^{-1}$, $L_v = 2.26 \times 10^6 \text{ Jkg}^{-1}$)

Do yourself

$$\text{Heat gained} = m_i L_i + (m_i + m_w) S_w \Delta \theta + m_c S_c \Delta \theta$$

$$= ?$$

$$\text{lost} = m_s L_v + m_s S_w (100 - 40)$$

$$= 2512000 m_s$$

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34

Q.13) What is the result of mixing 100 gm ice at 0 °C into 100 gm of water at 20 °C in an iron vessel of mass 100 gm? ($S_i = 0.112 \text{ cal gm}^{-1}\text{K}^{-1}$, $L_i = 80 \text{ cal gm}^{-1}$)

Do yourself

$$\text{Heat lost by W \& I} = (ms\Delta\theta)_w + (ms\Delta\theta)_I$$

$$\text{Heat req. to convert ice to water at } 0^\circ\text{C} = mL_f$$

3/31/2025

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35

Some Short Question Answers

1. The specific latent heat of fusion of ice is $3.36 \times 10^5 \text{ J kg}^{-1}$. What does it mean?

- It means that $3.36 \times 10^5 \text{ J}$ amount of heat is required to change the state of 1 kg of ice into water at constant temperature.

2. Why do we feel cold when we spray perfume on our body?

- Perfume is highly volatile material so it evaporates very fast and as we know that evaporation is the cooling effect. So we feel cold when we spray perfume on our body.

3. When you come out of swimming pool, you feel cold, why?

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4. Why is latent heat of vaporization of a material greater than that latent heat of fusion?

- When a liquid changes $L_v > L_f$ into a gas, there is a large increase in the volume. Therefore, a large amount of work has to be done against the surrounding atmosphere which comes in the form of latent heat. That is why, latent heat of vaporization of a material is greater than that of latent heat of fusion.

3/31/2025

Department Of Physics

36

5. A person gets a more severe burn by steam at 100 °C than by boiling water at 100 °C. Why?

- The specific latent heat of vaporization of water is 540 cal/gm. This means that when 540 cal is added to 1 gm of water at 100 °C then it is converted into steam at 100 °C. So every gm of steam contains 540 cal of more heat than 1 gm of water at 100 °C. So steam at 100 °C will produce more heat than water at 100 °C. So burn caused by steam will be more painful than that caused by boiling water.

6. Explain why water remains cool in earthen pot in summer.

- The earthen pot consists of a large number of pores. Through these pores, water comes out and evaporation takes place all over the pot. For the evaporation process, latent heat is absorbed from the water inside the pot. So the water inside the pot loses heat. This makes the pot cooler.

7. In summer, why do dogs hang their tongues while lazing under trees?

- Dogs do not have sweat glands. So they hang their tongues in order to expose a surface to the evaporation. The evaporation occurs on the tongue which gives the cooling.

3/31/2025

Department Of Physics

37

8. Why is it harmful to wear wet clothes on body?

- This is because the water on the wet clothes absorbs necessary latent heat from the body and then it evaporates. Due to evaporation process, cooling effect occurs. As a result, the body likely to catch cold.

9. A hole is made in an ice block at 0 °C and some water is filled in it. Will the water freeze?

- Water kept in the hole loses heat to the ice and cools down to 0 °C. Here, both ice and water are at same temperature so that water can not give out latent heat to the ice and hence it does not freeze.

10. Can water be made to boil without heating it?

- Yes, boiling point can be decreased by reducing pressure on the surface of water. If the pressure is greatly reduced so that the boiling point becomes equal to or less than the room temperature, then water boils without heating it.

3/31/2025

Department Of Physics

38

11. We feel cold during the melting of snow than during the snow fall, why?

- Snow melts due to absorption of heat from its surroundings so temperature of atmosphere decreases and therefore we feel cold during the melting of snow than during the snow fall.

12. Why do you feel cold in mouth when you eat halls?

- When we put halls in mouth, one of its chemical constituents evaporates. The required heat for vaporization is taken from our mouth. Due to this reason our mouth loses heat and hence we feel cold.

13. Why is it difficult to walk on the surface of a frozen lake?

- Increase of pressure decreases the melting point of ice. When a person walks on the frozen ice, the pressure of his feet decreases the melting point of ice and consequently some ice melts to form water. This makes the surface slippery and hence it becomes difficult to walk on such a slippery surface.